Rebooting the Ocean Ecosystem:

A Concept Proposal for Integrated Microplastic Removal and Ocean Alkalinity Enhancement

Ocean Alkalinity Enhancement / Marine Microplastic Removal / TRWP Reduction

1 Nov 2026 Kenichi Miyauchi

Table of Contents

1. Introduction

2. Removal of Marine Plastic

Appendix A

3. Reduction of TRWP

Appendix B

4. Summary

1. Introduction

1-1. About This Proposal

This proposal presents an integrated approach to marine ecosystem restoration by combining marine microplastic removal, ocean alkalinity enhancement, and land-based measures to reduce tire and road wear particles(TRWP).

Why “restart” the ocean?

The ocean is Earth's largest sink for excess heat and CO_2.

Restoring this circulation system physically and scientifically is the most efficient and fundamental approach to mitigating global warming.

1-2. Two Critical Bottlenecks Hindering Carbon Absorption

1. Potential Impacts on the Marine Biological Carbon pump

Microplastics are present throughout the marine environment and may interact with plankton and marine food Webs.
These interactions could affect biological processes involved in the ocean carbon cycle, although their overall impact on the ocean's biological pump remains uncertain.

2. Ocean Acidification and Seawater Carbonate Chemistry

As atmospheric CO_2 concentrations rise, more CO_2 enters seawater, changing its carbonate chemistry and lowering seawater pH.
This process is known as ocean acidification.
Ocean acidification does not necessarily mean that total alkalinity is declining.
Ocean alkalinity enhancement is being investigated as a potential approach to increase the ocean's capacity to store additional carbon, subject to environmental assessment and scientific verification.

2. Removal of Marine Plastic

2-1. Estimated Marine Plastic Stocks (1/3)

- Estimated total marine plastic stock: approximately 5–10 million tonnes

(1) Floating plastic in surface and near-surface waters

Estimated at approximately 1–3 million tonnes

May contribute to light attenuation and affect feeding activities of marine organisms.

2-1. Estimated Marine Plastic Stocks (2/3)

(2) **Plastic deposited on the seafloor**

Estimated at approximately 2–5 million tonnes

Difficult to quantify; potential future target for recovery using underwater exploration robots.

One of the least visible components of marine plastic pollution.

2-1. Estimated Marine Plastic Stocks (3/3)

(3) **Nanoplastics (poorly quantified component)**

Estimated at approximately 1–2 million tonnes

Can be generated through the fragmentation of larger plastics by UV exposure and mechanical abrasion.

Difficult to detect and quantify using current monitoring methods.

May interact with marine organisms and potentially affect biological carbon cycling.

2-2-1. Potential Restoration of Ocean CO_2 Uptake Capacity

The following measures are proposed to investigate the potential for supporting the ocean's natural CO_2 uptake capacity:

(1) Remove microplastics from surface and near-surface waters.

(2) Reduce light attenuation caused by suspended plastics, potentially improving light availability for plankton.

(3) Enhance seawater alkalinity under carefully controlled conditions.

2-2-2. Potential Restoration of Ocean CO_2 Uptake Capacity

- Operate using vessels in the 20–40 m class.
- Collect seawater from the surface layer (approximately 1 m) and shallow layer (approximately 2–5 m).
- Filter seawater using ultrafiltration (UF) membranes.
  - Remove suspended microplastics and particles.
  - Apply electrocoagulation as a supplementary treatment process.
- Enhance the alkalinity of treated seawater through controlled lime addition.
- Target pH and total alkalinity levels will be determined through site-specific modelling, laboratory testing, and environmental assessment.

2-2-2. Potential Restoration of Ocean CO_2 Uptake Capacity

- Target pH and total alkalinity levels will be determined through site-specific modelling, laboratory testing, and environmental assessment.
  - ✓ Monitor pH, total alkalinity, temperature, and other relevant water quality parameters before and after treatment.
  - ✓ Conduct environmental monitoring to assess potential impacts on marine organisms.
- Release the treated seawater through a controlled dispersion system at the aft section of the vessel.

- No operations will be conducted in the territorial waters of other countries.

  (Japan’s activities are limited to its own maritime zone.)
- Renewable energy sources are preferred for powering filtration units, pumps, and monitoring equipment.

- Biodiesel or other lower-carbon marine fuels are preferred, subject to availability and environmental assessment.

2-2-3. Definition of MA Vessel

Microplastic & Alkalinization Vessel (MA Vessel)

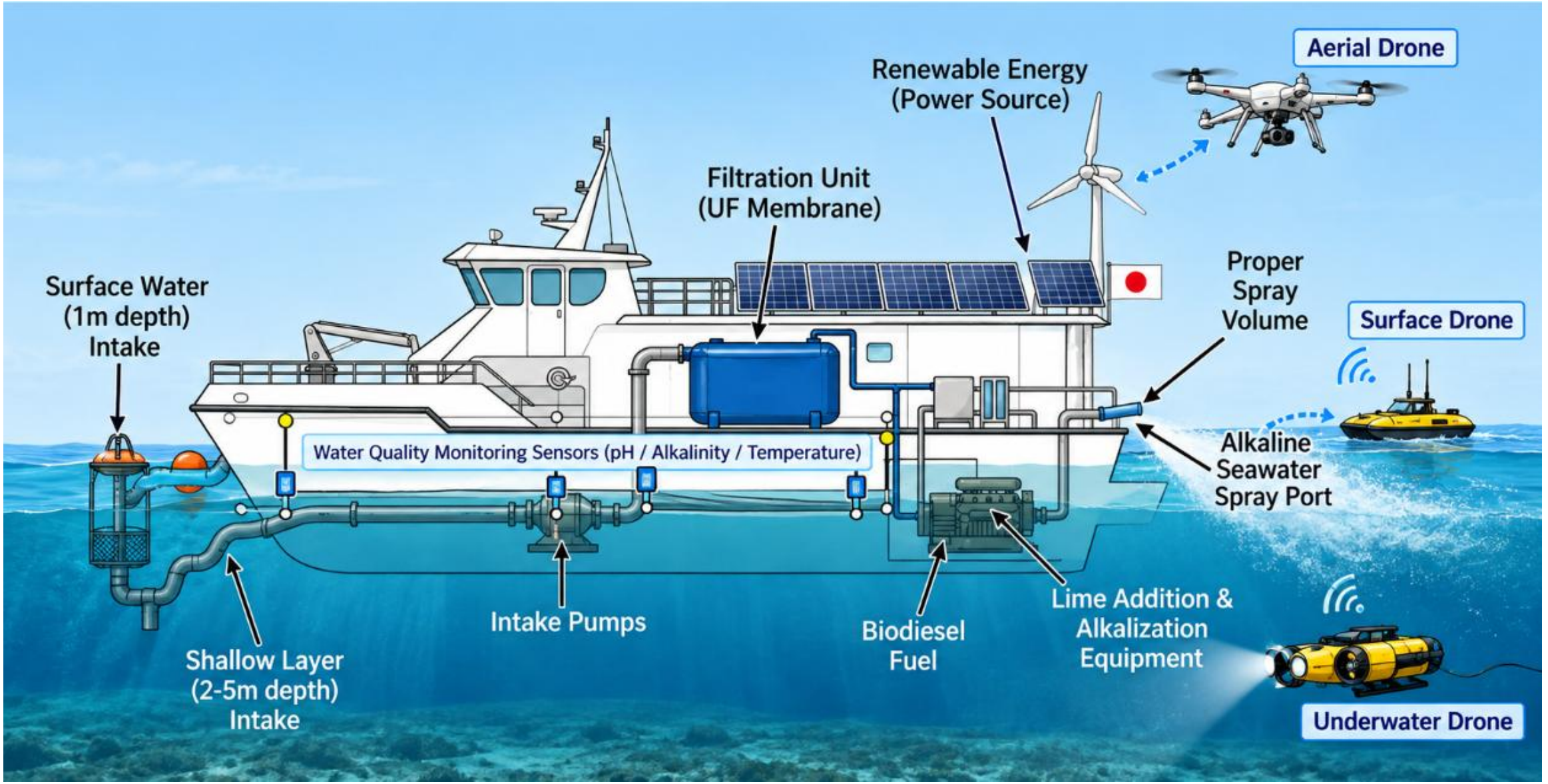
A specialized vessel designed to:

- **Remove microplastics** using ultrafiltration and electrocoagulation
- **Alkalinize treated seawater using a controlled amount of lime**
- **Disperse treated seawater** to investigate its potential contribution to ocean carbon uptake.

  Operate strictly within national maritime zones
- Use **renewable energy** and **biodiesel** to minimize emissions
- Monitor seawater quality, including pH, total alkalinity, and temperature.
- Coordinate with aerial, surface, and underwater drones for ocean observation and environmental monitoring.

The MA Vessel is the core operational unit integrating marine microplastic removal, ocean alkalinity enhancement, and environmental monitoring.

2-2-3. MA Vessel – Integrated Treatment and Monitoring System



2-3-1. Assumptions

- Microplastic concentration varies by region.
- Daily collection capacity per vessel: **5–20 kg/day**.
- Assumed annual operating period: 180 days, allowing for weather-related downtime.

Estimated Annual Collection per Vessel

Daily Collection Capacity	Annual Operating Days	Annual Collection Capacity
5 kg/day	180 days	900 kg/year
10 kg/day	180 days	1,800 kg/year
12.5 kg/day(avg.)	180 days	2,250 kg/year
15 kg/day	180 days	2,700 kg/year
20 kg/day	180 days	3,600 kg/year

Calculation based on an average collection rate of 12.5 kg/day per vessel.

2-3-2. Expected Impact of Multi-Vessel Deployment

- **Estimated Annual Removal Capacity with 100–1,000 Vessels:**

  100 vessels: 225 tonnes/year (depending on regional microplastic density)

  1,000 vessels: 2,250 tonnes/year (depending on regional microplastic density)

- **Estimated Cumulative Removal Over 10 Years (Based on 1,000 Vessels):**

  22,500 tonnes

  (assuming continuous operation at the estimated annual removal capacity)

- **Potential reduction of light attenuation and impacts on feeding activities caused by suspended microplastics**

2-3-3. Operation Plan for the MA Vessel

Daily Operating Parameters per MA Vessel

Item	Details
Operating Speed (during collection and dispersion)	4 knots (approx. 7 km/h)
Daily Microplastic Collection Capacity	10–15 kg/day
Alkalinized Seawater Volume	300–600 m^3/day
Estimated Sea Surface Area Covered	2–4 km^2/day
Seawater Dispersion Rate per Unit Area	0.15 L/m^2

2-3-4. Daily Operation Schedule for the MA Vessel

Daily Operation Schedule

Item	Details
Departure and transit to offshore operating area	1.0 h
Morning operations	2.5 h (with short breaks as needed)
Lunch break	1.0 h
Afternoon operations	2.5 h (with short breaks as needed)
Return transit to port	1.0 h
Cleanup & inspection	0.5 h
Total	8.5 h (approx. 8–9 hours)
Deteriorating Weather	Operations suspended; immediate withdrawal

2-3-5. Potential Ocean Operating Regions

- Potential Partner Countries and Coastal Areas

Ocean Region	Potential Partner Countries
North Pacific	United States, Canada, Russia, Japan
South Pacific	Chile, New Zealand
North Atlantic	United States, Canada
Eastern North Atlantic / Mediterranean	United Kingdom, France, Spain, Portugal, Ireland, Norway, Italy, Greece, Türkiye, North African coastal countries
South Atlantic	Brazil, Argentina
Indian Ocean	India, Sri Lanka, East African nations, Australia
Western Pacific - East China Sea - Yellow Sea - South China Sea	China, South Korea, Taiwan, Southeast Asian countries

- Landlocked countries that generate plastic waste may also be invited to participate in international cooperation efforts.

2-3-6. Expected Long-Term Impact

If these activities are continued and their effectiveness is demonstrated through scientific validation, they may help support the ocean’s natural carbon uptake processes over the long term.

The scale of any potential effect would need to be established through long-term monitoring and independent scientific assessment.

Illustrative, preliminary estimate: 50,000–200,000 t-CO_2/year

(subject to independent scientific validation and field measurements)

Appendix A. Complementary Marine Plastic Recovery Measures (Outside the Scope of This Proposal)

The following complementary measures may be considered alongside MA Vessel operations:

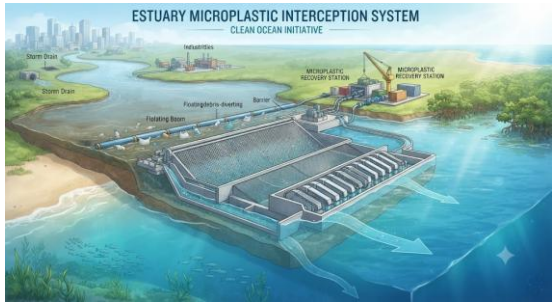
(1) Removal at Major Accumulation Zones

- North Pacific Gyre (Great Pacific Garbage Patch)
- South Pacific Gyre
- North Atlantic Gyre
- South Atlantic Gyre
- Indian Ocean Gyre

(2) Measures to Intercept Plastic Waste at River Mouths

(3) Seafloor Plastic Recovery Using Underwater Exploration Robots (Future Option)







3. Reduction of TRWP

3-1. Why Reduce **TRWP**

TRWP (Tire and Road Wear Particles) can contain harmful chemical substances and contribute to microplastic pollution.

Reducing **TRWP** at the source can help prevent these particles from entering waterways, accumulating in marine ecosystems, and causing long-term environmental impacts.

3-2. Key reasons for **TRWP** Reduction

Major microplastic source: **TRWP** are a significant source of microplastic pollution from land-based activities.

Toxic chemical content: **TRWP** can contain zinc and PAHs, while 6PPD-quinone is a transformation product associated with tire wear and can be toxic to aquatic organisms.

Biological ingestion: Their small particle size makes them easily ingested by fish, shellfish, and other marine life.

Persistent accumulation: **TRWP** can accumulate in sediments and may persist in the environment.

Source-level effectiveness: Reducing **TRWP** at the source can help prevent them from entering drainage systems and waterways, reducing the need for downstream cleanup.

3-3. TRWP Reduction Methods

(1) We propose installing a cyclone-type dust collector at the rear of vehicles to capture TRWP generated during driving.

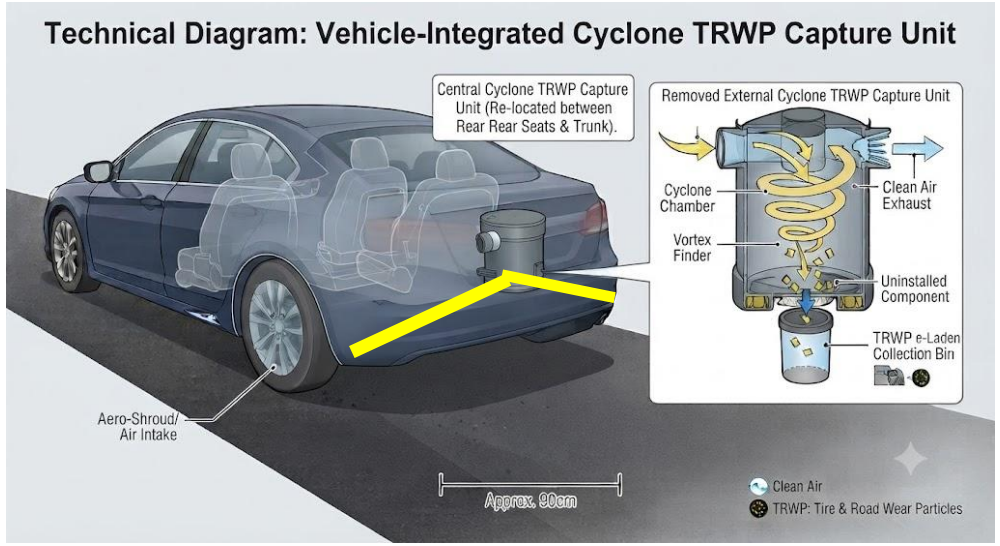
(2) We propose installing cyclone-type dust collectors near major intersections on arterial roads to capture TRWP from road surfaces.

(3) We propose measures to prevent TRWP washed from road surfaces into roadside gutters during heavy rain from entering rivers.

3-4-1. Vehicle-Mounted TRWP Capture Unit (Conceptual Design)

- A TRWP capture unit is designed to be mounted at the rear of the vehicle.
  - The unit uses a cyclone-type separator to capture airborne TRWP from the airflow.
  - Suction power is adjusted according to vehicle speed:
    higher suction at higher speeds and lower suction at lower speeds.
  - The system may be deactivated during rainfall, when airborne resuspension is expected to decrease and runoff becomes the primary transport pathway.
  - A clogging detection function is included to support stable operation.

- The unit is designed to be waterproof and mud-resistant for durability.



Conceptual proposal —
aerodynamic and field testing required

3-4-2. **Deployment of TRWP Capture Units on Major Roadways**

We propose installing TRWP capture units along major roadways, particularly near intersections and along median strips.

- Highways
- Primary roads
- Secondary roads

3-4-3. Preventing TRWP from Reaching Rivers and the Ocean via Roadside Runoff

- TRWP can be captured using microfiltration (MF) membranes before roadside runoff reaches rivers.

- Design capacity should be determined based on local extreme rainfall conditions.

  An illustrative design rainfall intensity of 150 mm/h is assumed.

3-5. Power Source for **TRWP** Capture Units

- **TRWP** capture units installed near intersections on major arterial roads could be powered by locally generated solar electricity combined with energy storage, subject to site-specific energy-demand and feasibility assessments.

- Perovskite photovoltaic modules or coatings could be integrated into suitable infrastructure, such as traffic signal poles, road signs, and pedestrian overpasses, where installation is technically feasible.

- Energy storage may use sodium-ion batteries or other suitable non-lithium technologies, selected based on safety, performance, availability, and lifecycle assessment.

Appendix B. “TRWP Reduction Methods” 1/3 (Outside This Proposal)

Tire-side

(1) Low-wear tires (Europe’s ongoing initiative)

- Adoption of tires designed to reduce abrasion.
- The EU is introducing tyre abrasion limits under the Euro 7 Regulation, with phased implementation starting in 2028.

Road-side / Infrastructure Measures

(2) Road surface optimization

- Road surface characteristics and pavement materials can influence TRWP generation. Further research is needed to identify suitable low-wear pavement designs.

Appendix B. “TRWP Reduction Methods” 2/3 (Outside This Proposal)

(3) Improved drainage systems

- Side-gutter filters or traps to capture TRWP before entering waterways.
- Simple screen filters (e.g., 10–100 μm) may help capture TRWP, depending on particle size distribution and hydraulic conditions.
- Similar measures have already been introduced or are being investigated in parts of Europe.

(Also incorporated into this proposal as a measure to prevent TRWP from entering rivers and the ocean.)

(4) Street sweeping with fine-particle capture

- High-performance sweepers designed to collect fine particles. Actual capture efficiency depends on equipment specifications and operating conditions.

Appendix B. “TRWP Reduction Methods” 3/3 (Outside This Proposal)

(5) Traffic flow optimization

- Reducing stop-and-go traffic can help reduce tire wear.
- Smoother traffic patterns may help reduce TRWP generation.

(6) Speed management

- Higher driving speeds may increase tire wear and TRWP generation.
- Some EU cities have introduced lower speed limits in environmentally sensitive areas.

(7) Roadside vegetation strips

- Vegetation strips may help capture or retain some airborne particles.
- Studies have investigated the potential of roadside vegetation to reduce particulate matter dispersion.

A combined portfolio of measures is likely to be more effective than any single intervention.

4. Summary

Climate and Ocean Recovery Actions

- Capture and remove microplastics from the ocean surface and upper layers.
- Increase ocean alkalinity through controlled treatment, with pH targets determined by site-specific scientific assessment.
- Help prevent TRWP from entering rivers and the ocean.

→ Aims to support the recovery of the ocean's natural CO_2 absorption capacity.

→ May contribute to reducing excess heat accumulation in the ocean over the long term.

→ In the long term, reducing atmospheric CO_2 can contribute to a reduction in radiative forcing and global warming.

The timescale and magnitude of this effect require scientific modelling.

We cannot wait until global CO_2 emissions reach net-zero in 2050.

Early action is essential.

Open Knowledge Statement

This document is published as Open Knowledge / Open Documentation under the MIT License and CC-BY.

Anyone may freely use, modify, and redistribute it.

I am entrusting this to a promising young person.

1 Nov 2026 (Updated)

24 July 2026 (First edition)

Kenichi Miyauchi

mailto:miyauchi [dot] kenichi [at] outlook [dot] com

■ Revision History

Date	Changes	Notes
24 Jul 2026	First edition published	
01 Nov 2026	Updated TRWP reduction methods and summary; refined technical descriptions and scientific qualifications.	